

# An Interactive Display of Synthetic Jet Actuator with Real-Time Actuation Control

Zhanghe(Nomi) Wang<sup>1</sup>, Stella Gerlock<sup>2</sup>,  
Chukwudum Eluchie<sup>3</sup>, Skinder Ali Dar<sup>3</sup>, Cong Wang<sup>3</sup>  
Mechanical Engineering  
IIHR – Hydrosience and Engineering

## Motivation

- **Engineering application:** Synthetic jet (SJ) has many engineering applications. An example is microrobots designed for targeted drug delivery. They use vortex-based propulsion to maneuver efficiently in confined environments, offering significant potential in biomedical applications.(Fig. 1)
- **Interactive education tool:** Experience how synthetic jets work and how their performance depends on frequency.

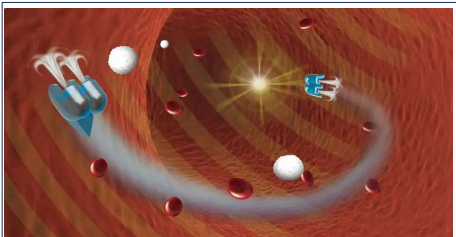


Fig.1 Microrobot swimmers propelled through blood vessels via vortex-based propulsion, inspired by synthetic jet concept. Adapted from Technology Networks (<https://www.technologynetworks.com/drug-discovery/news/micro-robot-swimmers-powered-by-ultrasound-353922>).

## Challenges:

- **Timing synchronization of rotary encoder:** Early tests showed erratic frequency jumps because each encoder detent produces four fast edge transitions, with mechanical bounce adding unwanted signals. Our initial polling routine missed or duplicated pulses. **To fix this issue, we implemented quadrature state-machine** (See Fig.5)
- **Illumination for flow visualization:** Standard LEDs lacked sufficient intensity to visualize particle motion in water. We addressed this by employing a green laser paired with a cylindrical lens to generate a high-intensity laser sheet, effectively illuminating neutrally buoyant particles for improved visualization.

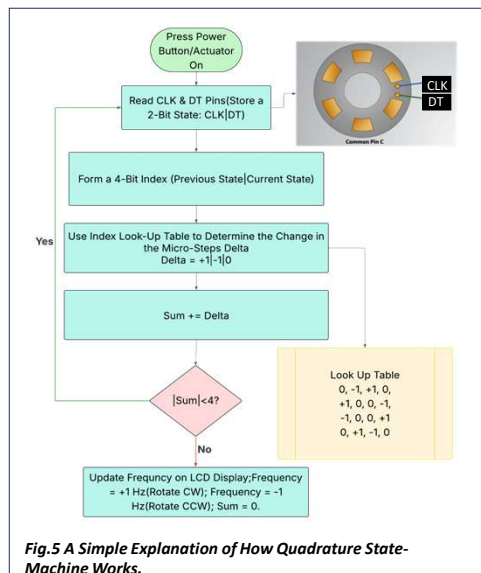


Fig.5 A Simple Explanation of How Quadrature State-Machine Works.

## System Design

- **Hardware Overview:** ESP32 microcontroller, rotary encoder, LCD display, push button, amplifier, voice coil actuator, laser and cylindrical lens. (See Fig.2)
- **Actuator Mechanism:** The actuator consists of a sealed cavity with an orifice on one side and a flexible membrane on the opposite side. Driven by the ESP32's sinusoidal output, the membrane vibrates, creating cyclic suction and expulsion of water through the orifice. At optimal frequencies, this forms distinct vortex rings characteristic of synthetic jets. (See Fig.3)
- **Circuit Design:** See Fig 4.

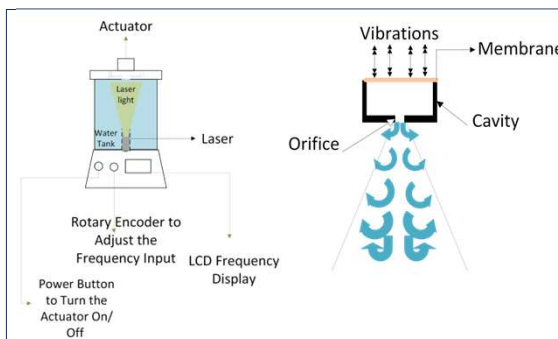


Fig.2 (Left) System Overview Diagram of the Interactive Synthetic Jet Display  
Fig.3(Right) Schematic Illustration of Synthetic Jet(SJ)

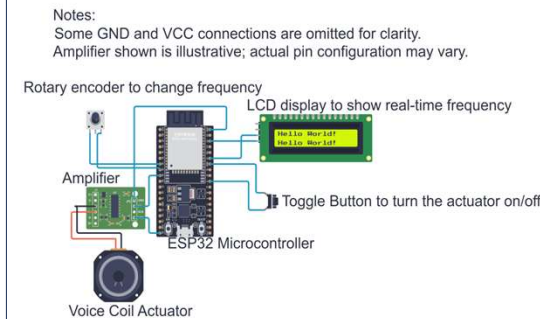


Fig.4 Control Circuit Schematics for the Synthetic Jet Actuator System

## Conclusions & Results

- **Frequency-Dependent Flow Visualization:** Testing across frequencies from 1 to 20 Hz demonstrated clearer and more defined vortex-ring structures around 5Hz. Although the actuator's small scale limited visual clarity, we successfully demonstrated frequency-dependent behavior.
- **Educational Impact:** The interactive setup enabled intuitive exploration and understanding of fluid mechanics phenomena without prior knowledge.

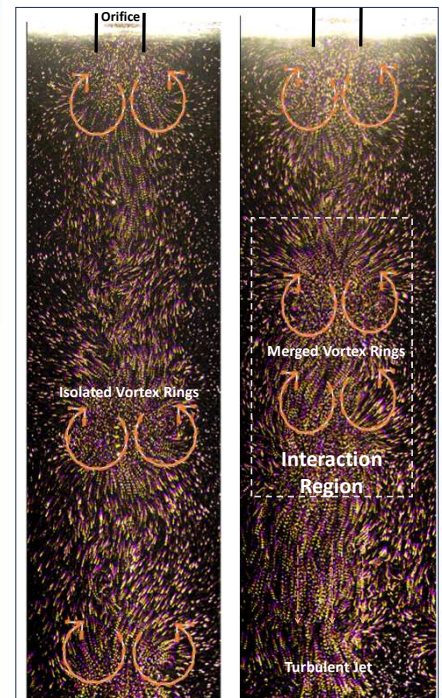


Fig.6 Flow Pathline Visualization of Vortical Structures in Synthetic Jet (Left - 5Hz; Right - 10Hz)

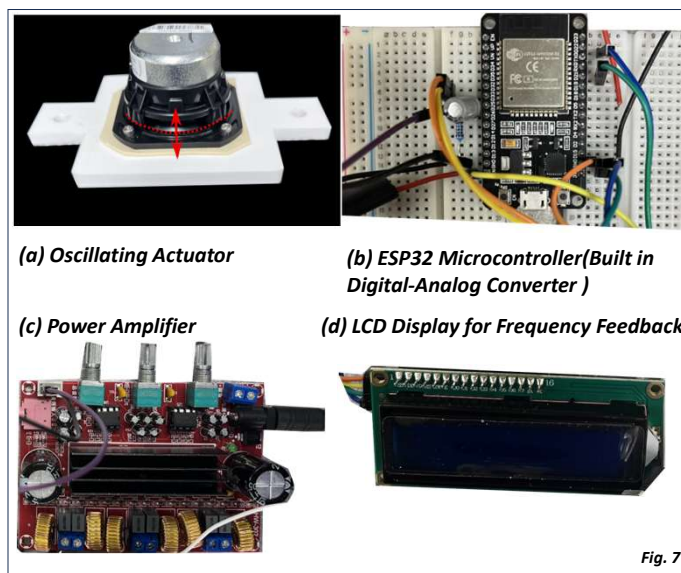


Fig. 7

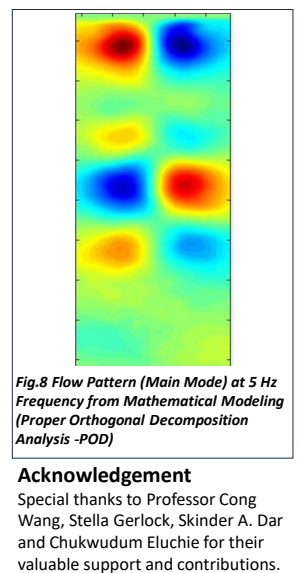


Fig.8 Flow Pattern (Main Mode) at 5 Hz Frequency from Mathematical Modeling (Proper Orthogonal Decomposition Analysis -POD)

## Acknowledgement

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